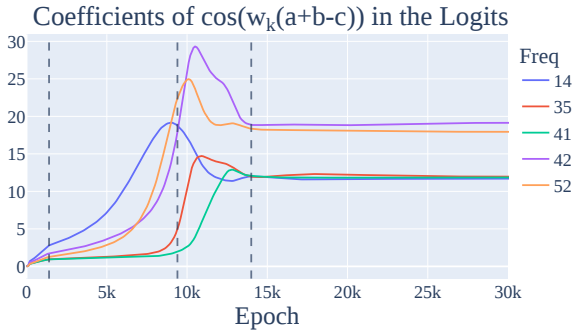

Figure 14: The coefficients of $\cos(w(a+b-c))$ in the logits over the model's training. As with the metrics in the paper, this shows a nice interpolation and growth of each cosine term.

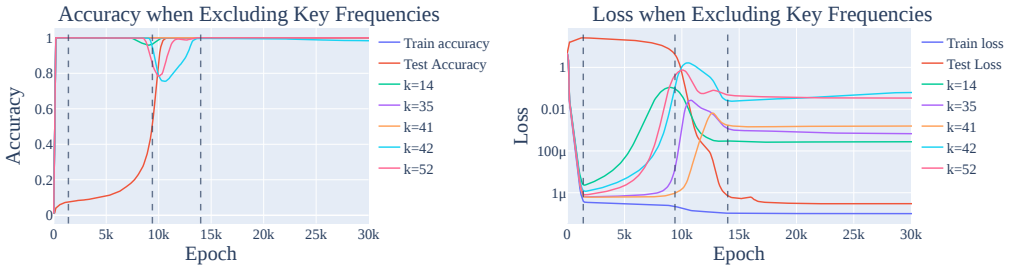

Figure 15: The excluded accuracy (left) and loss (right) if we exclude each of the five key frequencies for our mainline model. As with the excluded loss results in Section 5.1, this shows that the model interpolates between memorising and generalising.

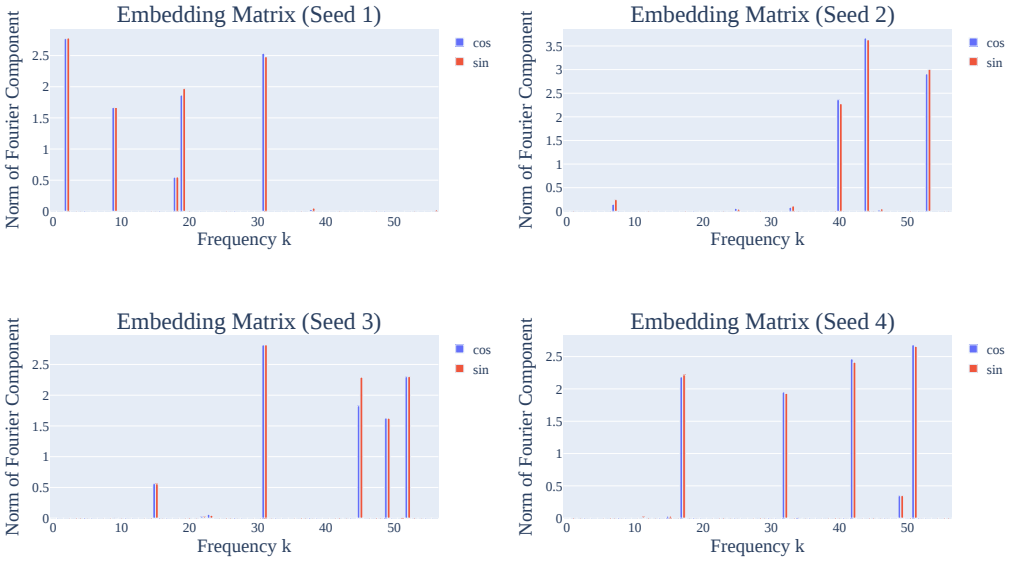

Figure 16: The norms of the Fourier components in the embedding matrix W_E for each of four other random seeds for the original (1 layer) architecture. As discussed in Section 4.1 and Appendix C.2.1, the sparsity of W_E in the Fourier basis is evidence that the network is operating in a Fourier basis.

W_L Component	Fourier components of u_k^TMLP(a,b) or v_k^TMLP(a,b)	FVE
$\cos(w_2c)$	$147.4\cos(w_2a)\cos(w_2b) - 145.8\sin(w_2a)\sin(w_2b) \approx 146.6\cos(w_2(a+b))$	99.2%
$\sin(w_2c)$	$145.5\cos(w_2a)\sin(w_2b) + 145.6\sin(w_2a)\cos(w_2b) \approx 145.5\sin(w_2(a+b))$	99.1%
$\cos(w_9c)$	$49.3\cos(w_9a)\cos(w_9b) - 48.0\sin(w_9a)\sin(w_9b) \approx 48.6\cos(w_9(a+b))$	96.4%
$\sin(w_9c)$	$48.6\cos(w_9a)\sin(w_9b) + 48.5\sin(w_9a)\cos(w_9b) \approx 48.5\sin(w_9(a+b))$	96.7%
$\cos(w_{19}c)$	$58.0\cos(w_{19}a)\cos(w_{19}b) - 58.3\sin(w_{19}a)\sin(w_{19}b) \approx 58.2\cos(w_{19}(a+b))$	95.4%
$\sin(w_{19}c)$	$59.3\cos(w_{19}a)\sin(w_{19}b) + 59.4\sin(w_{19}a)\cos(w_{19}b) \approx 59.4\sin(w_{19}(a+b))$	93.9%
$\cos(w_{31}c)$	$94.4\cos(w_{31}a)\cos(w_{31}b) - 96.4\sin(w_{31}a)\sin(w_{31}b) \approx 95.4\cos(w_{31}(a+b))$	98.4%
$\sin(w_{31}c)$	$97.2\cos(w_{31}a)\sin(w_{31}b) + 97.1\sin(w_{31}a)\cos(w_{31}b) \approx 97.2\sin(w_{31}(a+b))$	98.7%

(a) Seed 1

W_L Component	Fourier components of u_k^TMLP(a,b) or v_k^TMLP(a,b)	FVE
$\cos(w_{40}c)$	$97.0\cos(w_{40}a)\cos(w_{40}b) - 99.4\sin(w_{40}a)\sin(w_{40}b) \approx 98.2\cos(w_{40}(a+b))$	97.3%
$\sin(w_{40}c)$	$81.3\cos(w_{40}a)\sin(w_{40}b) + 81.3\sin(w_{40}a)\cos(w_{40}b) \approx 81.3\sin(w_{40}(a+b))$	92.7%
$\cos(w_{44}c)$	$309.1\cos(w_{44}a)\cos(w_{44}b) - 338.7\sin(w_{44}a)\sin(w_{44}b) \approx 323.9\cos(w_{44}(a+b))$	98.5%
$\sin(w_{44}c)$	$327.3\cos(w_{44}a)\sin(w_{44}b) + 327.2\sin(w_{44}a)\cos(w_{44}b) \approx 327.3\sin(w_{44}(a+b))$	98.9%
$\cos(w_{53}c)$	$192.1\cos(w_{53}a)\cos(w_{53}b) - 192.2\sin(w_{53}a)\sin(w_{53}b) \approx 192.1\cos(w_{53}(a+b))$	97.3%
$\sin(w_{53}c)$	$166.7\cos(w_{53}a)\sin(w_{53}b) + 166.8\sin(w_{53}a)\cos(w_{53}b) \approx 166.8\sin(w_{53}(a+b))$	95.7%

(b) Seed 2

W_L Component	Fourier components of u_k^TMLP(a,b) or v_k^TMLP(a,b)	FVE
$\cos(w_{31}c)$	$156.1\cos(w_{31}a)\cos(w_{31}b) - 156.5\sin(w_{31}a)\sin(w_{31}b) \approx 156.3\cos(w_{31}(a+b))$	99.3%
$\sin(w_{31}c)$	$150.7\cos(w_{31}a)\sin(w_{31}b) + 150.7\sin(w_{31}a)\cos(w_{31}b) \approx 150.7\sin(w_{31}(a+b))$	98.9%
$\cos(w_{45}c)$	$72.5\cos(w_{45}a)\cos(w_{45}b) - 76.8\sin(w_{45}a)\sin(w_{45}b) \approx 74.6\cos(w_{45}(a+b))$	95.9%
$\sin(w_{45}c)$	$74.7\cos(w_{45}a)\sin(w_{45}b) + 74.6\sin(w_{45}a)\cos(w_{45}b) \approx 74.6\sin(w_{45}(a+b))$	96.6%
$\cos(w_{49}c)$	$45.9\cos(w_{49}a)\cos(w_{49}b) - 45.5\sin(w_{49}a)\sin(w_{49}b) \approx 45.7\cos(w_{49}(a+b))$	97.0%
$\sin(w_{49}c)$	$45.8\cos(w_{49}a)\sin(w_{49}b) + 45.8\sin(w_{49}a)\cos(w_{49}b) \approx 45.8\sin(w_{49}(a+b))$	96.9%
$\cos(w_{52}c)$	$71.6\cos(w_{52}a)\cos(w_{52}b) - 72.1\sin(w_{52}a)\sin(w_{52}b) \approx 71.9\cos(w_{52}(a+b))$	98.5%
$\sin(w_{52}c)$	$68.7\cos(w_{52}a)\sin(w_{52}b) + 68.7\sin(w_{52}a)\cos(w_{52}b) \approx 68.7\sin(w_{52}(a+b))$	97.9%

(c) Seed 3

W_L Component	Fourier components of u_k^TMLP(a,b) or v_k^TMLP(a,b)	FVE
$\cos(w_{17}c)$	$66.0\cos(w_{17}a)\cos(w_{17}b) - 63.5\sin(w_{17}a)\sin(w_{17}b) \approx 64.8\cos(w_{17}(a+b))$	96.4%
$\sin(w_{17}c)$	$66.4\cos(w_{17}a)\sin(w_{17}b) + 66.4\sin(w_{17}a)\cos(w_{17}b) \approx 66.4\sin(w_{17}(a+b))$	94.9%
$\cos(w_{32}c)$	$68.7\cos(w_{32}a)\cos(w_{32}b) - 68.4\sin(w_{32}a)\sin(w_{32}b) \approx 68.5\cos(w_{32}(a+b))$	96.2%
$\sin(w_{32}c)$	$68.0\cos(w_{32}a)\sin(w_{32}b) + 68.0\sin(w_{32}a)\cos(w_{32}b) \approx 68.0\sin(w_{32}(a+b))$	96.3%
$\cos(w_{42}c)$	$100.4\cos(w_{42}a)\cos(w_{42}b) - 96.0\sin(w_{42}a)\sin(w_{42}b) \approx 98.2\cos(w_{42}(a+b))$	97.9%
$\sin(w_{42}c)$	$100.2\cos(w_{42}a)\sin(w_{42}b) + 100.1\sin(w_{42}a)\cos(w_{42}b) \approx 100.1\sin(w_{42}(a+b))$	98.6%
$\cos(w_{51}c)$	$118.0\cos(w_{51}a)\cos(w_{51}b) - 116.2\sin(w_{51}a)\sin(w_{51}b) \approx 117.1\cos(w_{51}(a+b))$	99.0%
$\sin(w_{51}c)$	$114.3\cos(w_{51}a)\sin(w_{51}b) + 114.2\sin(w_{51}a)\cos(w_{51}b) \approx 114.2\sin(w_{51}(a+b))$	98.5%

(d) Seed 4

Table 3: For each of the directions in the neuron-logit map W_L of the final models from 4 other random seeds (Appendix C.2.1), we project the MLP activations in that direction then perform a Fourier transform. For brevity, we omit terms with coefficients less than 15% of the largest coefficient. We then compute the fraction of variance explained (FVE) if we replace the projection with a multiple of a single term of the form $\cos(w_k(a+b))$ or $\sin(w_k(a+b))$, and find that this is consistently close to 1.

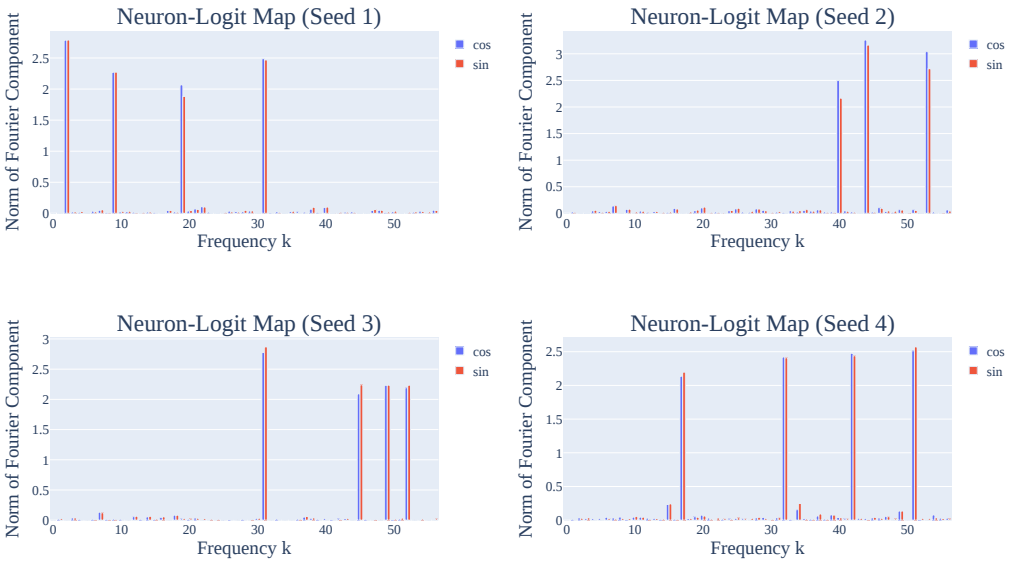

Figure 17: The norms of the direction corresponding to sine and cosine waves in the neuron-logit map weights W_L. As with the mainline model discussed in the main body and discussed in Appendix C.2.1, W_L is consistently sparse, providing is evidence that all four are operating in a Fourier basis.

Seed	Test Loss	Loss (Key frequencies removed)	Loss (All other frequencies removed)
1	$2.07 \cdot 10^{-7}$	$6.5 \cdot 10^{0}$	$5.7 \cdot 10^{-8}$
2	$2.1 \cdot 10^{-7}$	$1.1 \cdot 10^{1}$	$6.2 \cdot 10^{-8}$
3	$2.05 \cdot 10^{-7}$	$6.7 \cdot 10^{0}$	$5.5 \cdot 10^{-8}$
4	$2.33 \cdot 10^{-7}$	$6.8 \cdot 10^{0}$	$6.0 \cdot 10^{-8}$

Table 4: As discussed in Appendix C.2.1, ablating the key frequencies for each of the networks reduces performance to worse than chance, while ablating all other frequencies *improves* performance.

Confirming that the other seeds use the Fourier Multiplication Algorithm. In Figure 16, we show the norms of the Fourier components of the embedding matrix W_E for each of the 4 other random seeds. As with the mainline model, the matrices are sparse in the Fourier basis. In Figure 17, we show the norms of the Fourier components of the neuron-logit map W_L for the 4 other random seeds. The matrices are sparse in the Fourier basis, enabling us to identify 3 or 4 key frequencies for each of the seeds. Again, note that the specific frequencies differ by seed.

Using the key frequencies identified in the neuron-logit map, we repeat the experiment in Section 4.2, where we "read off" the MLP activations in the 6 or 8 directions corresponding to the key frequencies. As with our mainline model, this lets us identify the trigonometric identities for $\cos(w_k(a+b))$ and $\sin(w_k(a+b))$ being computed at the MLP layer. We confirm that the trigonometric identities are a good approximation by approximating the activations with a single term of the form $\cos(w_k(a+b))$ or $\sin(w_k(a+b))$—as with the mainline model, the fraction of variance explained is consistently close to 100%.

Next, we ablate the key frequencies from the logits as in Section 4.4 and report the results in Table 4. As with the mainline model, ablating all of the key frequencies reduces performance to worse than chance, while ablating everything *but* the key frequencies *improves* test performance.

Progress measures and grokking. Finally, we confirm the progress measure and grokking results from the mainline model on other runs with the same architecture. In Figure 18, we display the train, test, and restricted loss for each of the four other random seeds. In Figure 19, we display the Gini coefficients of the Fourier components of the embedding matrix W_E and the neuron-logit map W_L for each of the four other random seeds. The shape of the curves are very similar to those of the mainline model, allowing us to divide grokking on these models into the same three phases